SMART EV SECURITY SYSTEM USING GPS,GSM AND GEOFENCING

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ABSTRACT:

Electric Vehicles (EVs) need advanced security for protection against theft and unauthorized use. This paper presents an IoT-based EV security system using GPS, GSM, and dynamic geofencing. The system monitors the vehicle’s location, ignition, and movement in real time. If the vehicle crosses the geofence or detects unauthorized activity, GSM sends an SMS alert with the GPS location to the owner. A relay-based immobilization unit also allows remote disabling of the drive circuit through authenticated commands. The proposed system provides a low-cost, reliable, and scalable solution for EV security, theft prevention, and fleet monitoring.

PROBLEM STATEMENT :

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ELECTRIC VEHICLES REQUIRE EFFECTIVE SECURITY MECHANISMS TO PROTECT AGAINST:

- **Unauthorized vehicle movement**
- **Vehicle theft and unexpected displacement**
- **Lack of continuous location monitor**
- **Movement outside a permitted area**
- **Delayed detection of suspicious activity**
- **Absence of immediate local warning mechanisms**

PROBLEM STATEMENT SOLUTION :

HE PROJECT AIMS TO DEVELOP A LOW-POWER AND COST-EFFECTIVE EV SECURITY SYSTEM THAT:

- **Continuously monitors the vehicle's GPS location**
- **Establishes a user-defined home/authorized location**
- **Detects motion or presence using sensors**
- **Checks whether the vehicle has deviated from the permitted area**
- **Provides an immediate buzzer alert when a geofence violation is detected**
- **Displays important system information on an OLED**

OBJECTIVES :

- **GPS LOCATION MONITORING :**
- Obtain the real-time geographical position of the EV using the NEO-6M GPS module.

- **Home Location Initialization :**
- Use the push button to capture and store the current GPS position as the authorized/home location.

- **Geofencing :**
- Continuously compare the current GPS position with the stored home location.

- **Motion Detection :**
- Use the PIR sensor to detect movement or presence around the vehicle.

- **Magnetic/Orientation Monitoring :**
- Use the GY-271 QMC5883L magnetometer to identify magnetic-field or orientation changes associated with movement.

- **Local Alert :**
- Activate the buzzer when the geofence condition indicates unauthorized deviation.

- **Real-Time Display :**
- Display selected GPS and security values on the SSD1306 OLED.

LITERACTURE SURVEY :

S.no	Year	Paper Name	Methodology / Technology	Key contribution	Limitaion / gap
1	2024	**Position Verification in Connected Vehicles for Cyber Resilience Using Geofencing and Fuzzy Logic** *(IEEE Access, 2024)*	Geofencing, Fuzzy Logic, Vehicle Position Verification	Improves verification of vehicle position and addresses location-security issues in connected vehicles.	More focused on secure position verification in connected-vehicle environments than a low-cost embedded anti-theft prototype.
2	2024	**Literature Survey on Integrated Vehicle Tracking with Speed, Tilt, and Geofencing** *(IARJSET, 2024)*	ESP32, GPS, Speed Sensor, Tilt Sensor, Wi-Fi, Geofencing	Combines vehicle tracking with geofencing and sensor-based monitoring; tests geofence entry/exit detection.	Depends on IoT/Wi-Fi infrastructure and cloud-based monitoring; focuses on multiple vehicle parameters rather than local security alerting.
3	2024	**Automotive Cybersecurity: A Survey on Frameworks, Standards, and Testing and Monitoring Technologies** *(Sensors, MDPI, 2024)*	Automotive Cybersecurity, Monitoring, Standards, Testing	Reviews modern automotive cybersecurity challenges, monitoring technologies, standards and security frameworks.	Broad cybersecurity survey rather than a hardware-level vehicle security implementation.

4	2025	**Smart Vehicle Security System: Real-Time Tracking and Theft Prevention with IoT** *(IEEE I-PACT, 2025)*	GPS, GSM, IoT, Cloud (Firebase), Mobile Application	Provides real-time vehicle tracking, theft alerts, historical routes and remote engine-control functionality	Requires GSM/cloud infrastructure and multiple communication components, increasing system complexity.
5	2025	Penerapan Internet of Things (IoT) Untuk Pelacakan Keberadaan Kendaraan Bermotor Roda Dua (J-SKom, 2025)	ESP32, NEO-6M GPS, Geofencing, Buzzer, Kalman Filter, IoT (MQTT)	Uses ESP32 + NEO-6M with geofencing and buzzer; Kalman filter improves GPS accuracy.	Adds camera, MQTT, Telegram and web monitoring, making it more communication/software oriented than a simple local security prototype

RESEARCH GAP :

The literature survey shows that existing vehicle-security solutions mainly focus on GPS tracking, geofencing, IoT communication, theft detection and remote control. However, several limitations remain

External Communication: Dependence on GSM, Wi-Fi and cloud increases complexity.

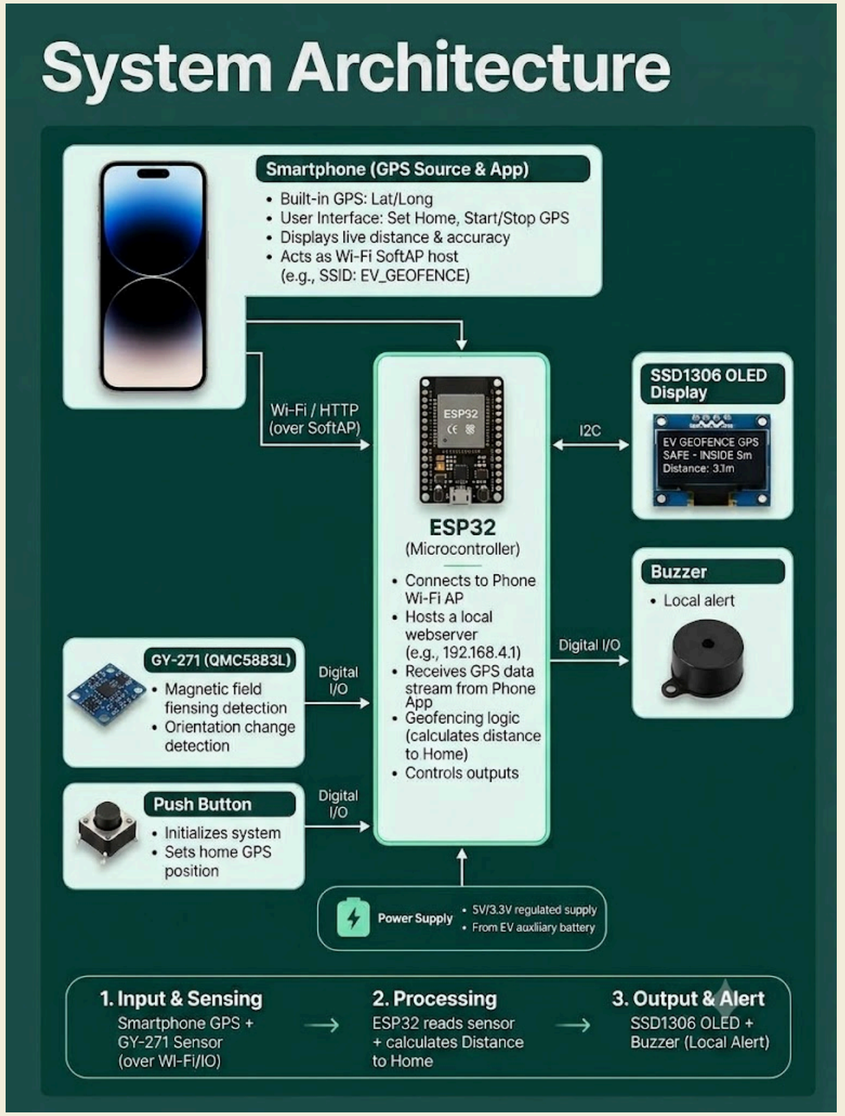
Local Response: Limited immediate local alerts for suspicious movement.

Multi-Sensor Integration: Limited combination of GPS, motion and magnetic sensing.

Monitoring Challenges: GPS accuracy can be affected in poor-coverage areas.

System Complexity: Additional modules increase cost, power use and complexity.

SYSTEM ARCHITECTURE



WORKING DEMONSTRATION :



WORKING METHODOLOGY :

INITIALIZE SYSTEM – ESP32 INITIALIZES ALL CONNECTED SENSORS AND OLED.

Set Home Location – Push button stores the current GPS position as the reference location.

Detect Movement – PIR detects motion/presence and GY-271 monitors magnetic/orientation changes.

Check GPS Deviation – ESP32 compares the current GPS position with the stored home position.

Geofence Decision – Determines whether the vehicle is inside or outside the permitted area.

Alert & Display – OLED shows system status and the buzzer activates when a geofence violation is detected.

SYSTEM OUTPUT & RESULT :

System output :

- DISPLAYS THE CURRENT GPS LOCATION ON THE OLED.
- Stores and displays the Home GPS location.
- PIR detects vehicle movement/presence.
- GY-271 detects magnetic/orientation changes.
- ESP32 performs the GPS deviation and geofence check.
- Buzzer provides an audible alert when the vehicle crosses the defined geofence.

Result:

NORMAL CONDITION: VEHICLE REMAINS WITHIN THE PERMITTED AREA → NO ALERT
Violation Condition: Vehicle moves beyond the permitted area → Geofence Violation → Buzzer ON

Conclusion & Future Scope :

Conclusion :

- Developed a low-cost EV security monitoring system using GPS and sensor-based movement detection.
- NEO-6M enables real-time location and geofencing.
- GY-271 provide additional movement information.
- OLED enables local system monitoring.
- Buzzer provides immediate local notification during a geofence violation.

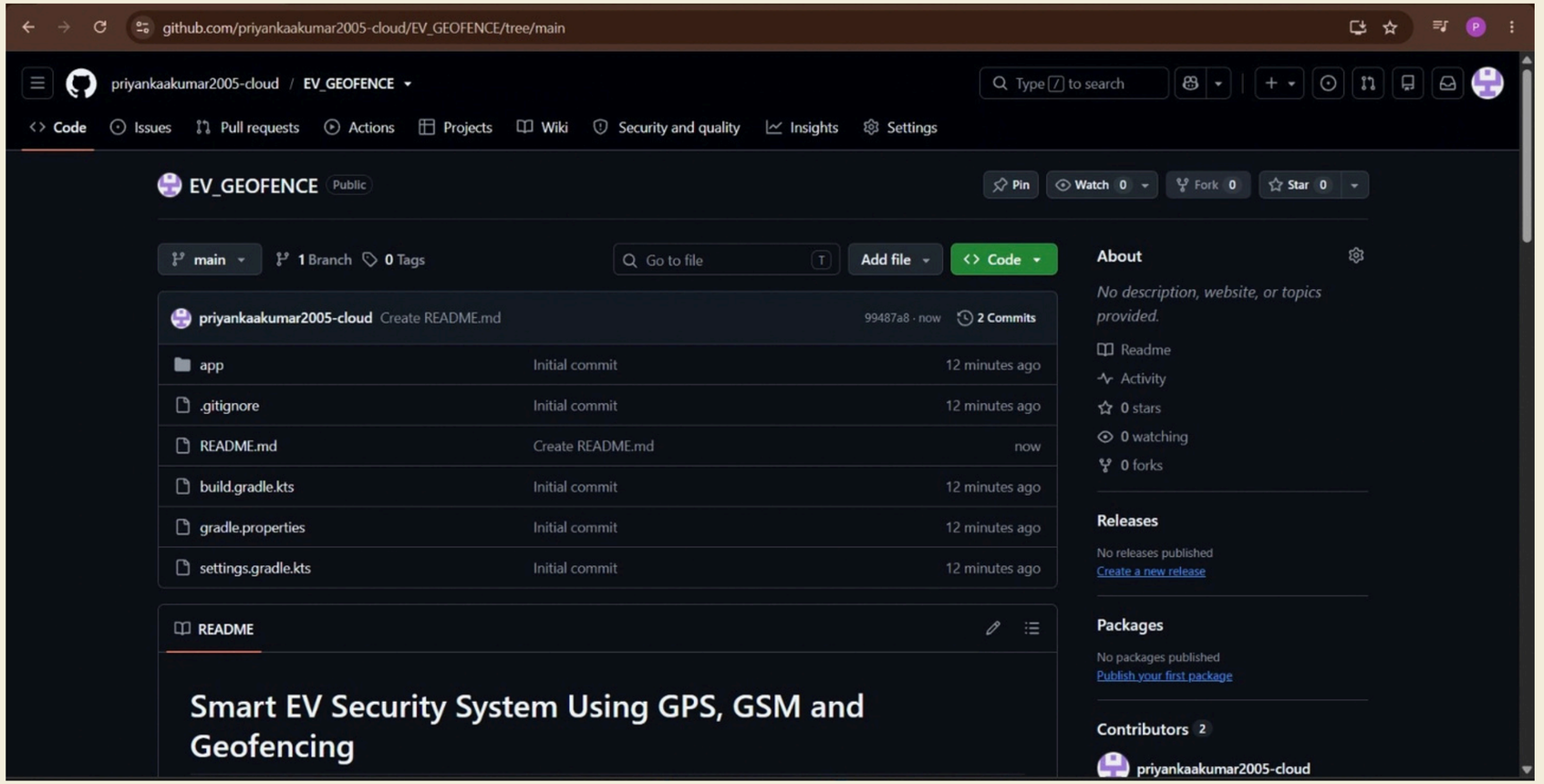
Future scope :

- GSM-based SMS alerts
- Mobile/Web-based live tracking
- Remote vehicle immobilization
- Cloud-based telemetry and monitoring
- AI/ML-based theft and abnormal-movement detection
- Improved geofencing

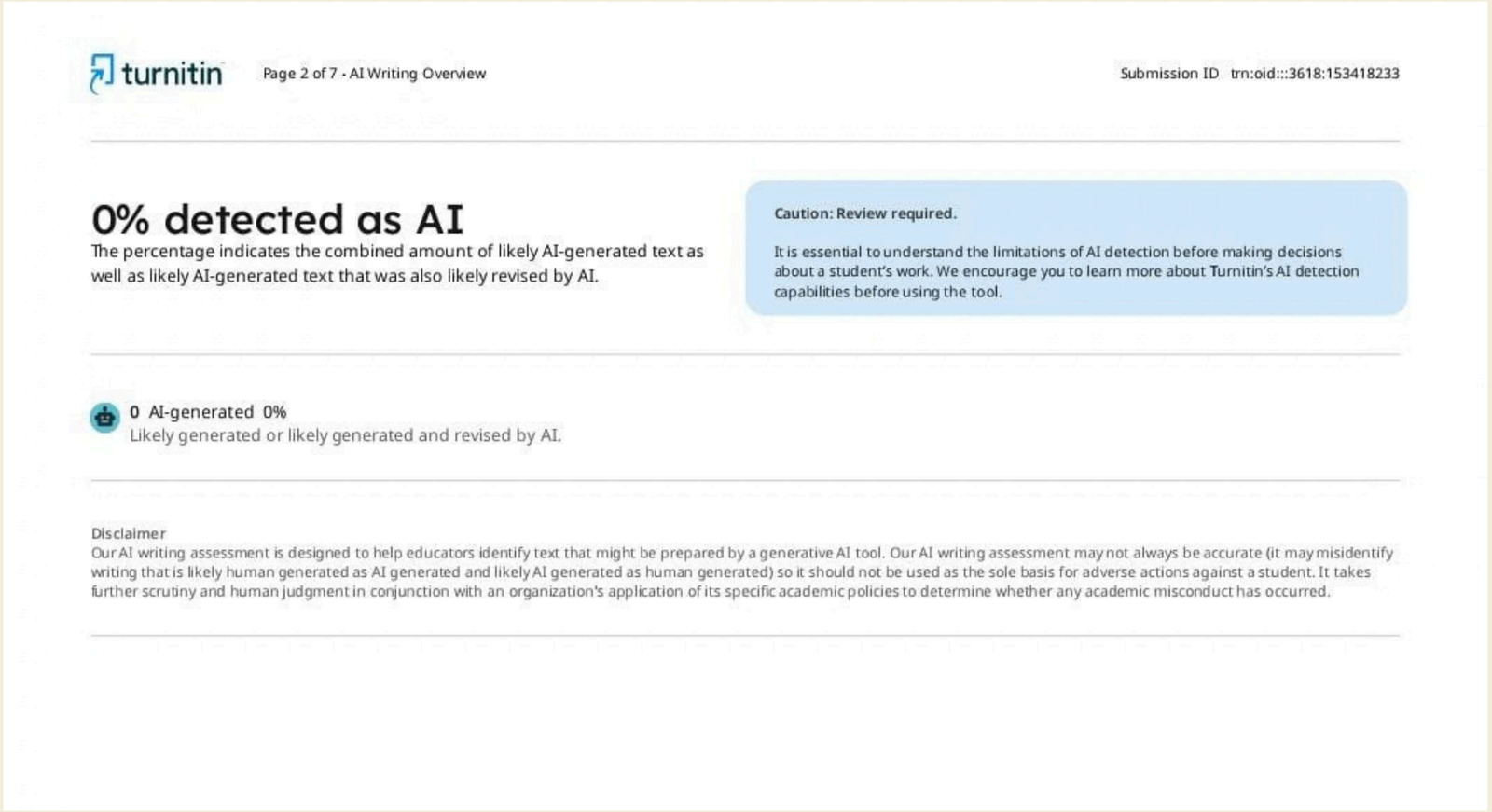
IEEE CONFRENCE PAPER TOPICS :

1. Introduction
2. Literature Survey
3. Proposed Methodology
4. Result And Discussions
5. Conclusion And Future Scope
6. References

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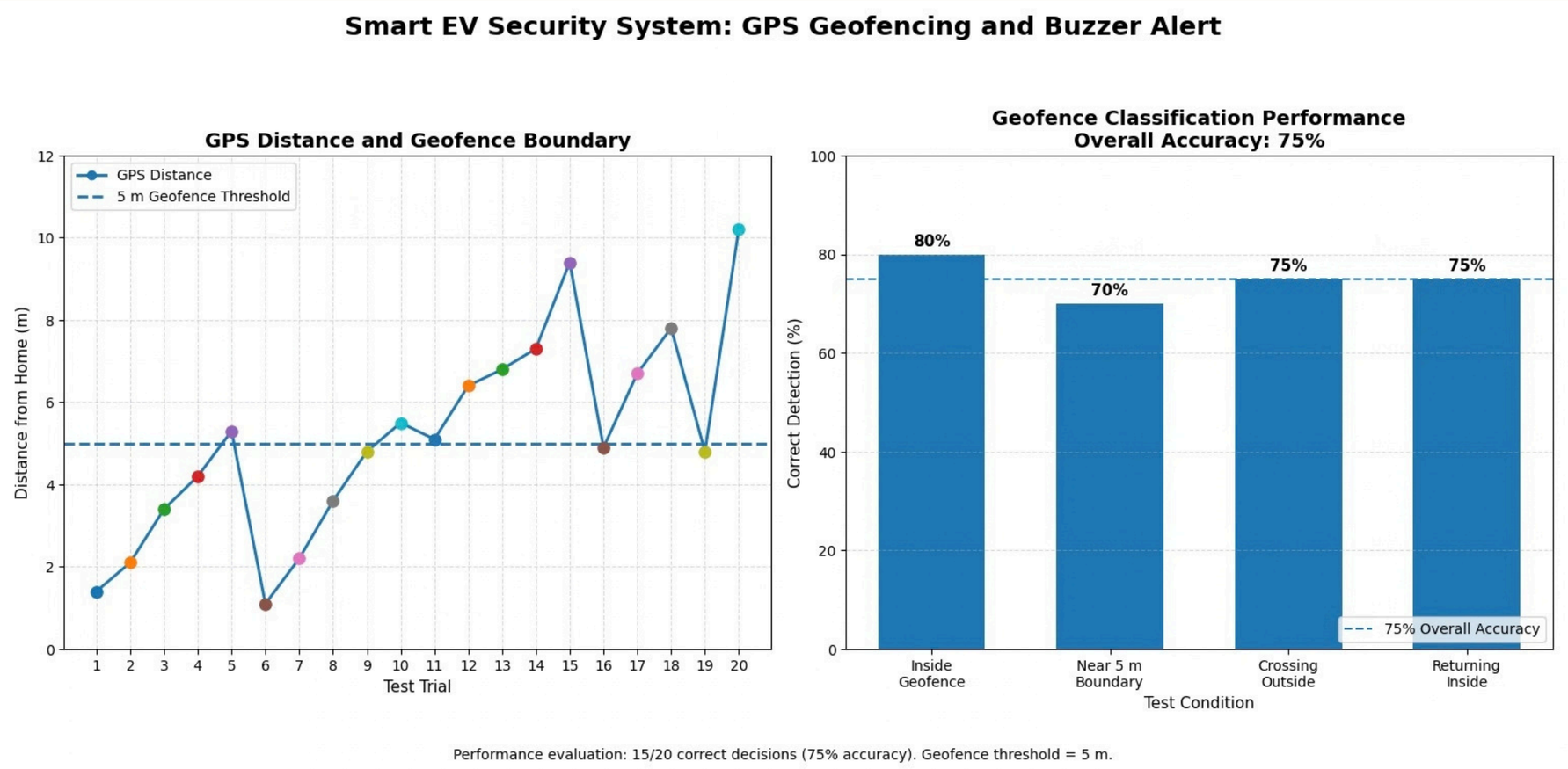
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PERFORMANCE ANALYSIS



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